MILITARY SPECIFICATION

PROJECTILE, 81MM, H.E., M374A1 METAL PARTS ASSEMBLY
(CARBON STEEL)

1. SCOPE

1.1 This specification covers Projectile 81MM, H.E., M374A1 Metal Parts Assembly (Carbon Steel).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-6868	- Inspection, Processes Magnetic Particle

STANDARDS

Federal

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection Sampling and Testing

Federal Test Method Standard No. 151 - Metals, Test Methods

FSC 1315

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2.1 (Cont'd)

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Qualification of Inspection Personnel (Magnetic Particle and Penetrant)
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedure and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

D10551951	- Projectile, 81MM, H.E., M374A1 Metal Parts Assembly (Carbon Steel)
IEL - 10551951	- Inspection Equipment List for Projectile, 81MM, H.E., M374A1 Metal Parts Assembly (Carbon Steel)

PUBLICATIONS

PA-PD-792	- Equipment, Testing or Measuring for Ammunition Items

(Copies of specifications, standards, drawings and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

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3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Mechanical properties.-All components shall meet the mechanical properties requirements as specified on the applicable drawings and shall be obtained by heat treating or cold working and stress relief.

3.2.1 Heat treatment.-When mechanical properties are achieved by heat treating, each projectile body shall be heated above the upper critical temperature, not to exceed 1600 degrees F., oil quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing.

3.2.2 Stress relieved.-When mechanical properties are achieved by cold shaping each cold worked body shall be stress relieved at a minimum temperature of 700 degrees F. and held at temperature for a minimum of 30 minutes. No work involving plastic flow of metal is permitted after final stress relief.

3.2.3 Hardness pattern.-Projectile bodies shall be sectioned and a hardness pattern shall be taken to determine the extent of hardness variation, location of hardness transition zones, absolute hardness values, etc, occuring within any one projectile body produced by the manufacturing process.

3.3 Metal defects.

3.3.1 Body.-The body, including the cavity shall be free from cracks, splits, bursts, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

3.3.2 Body cavity.-The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.3.3 Visual standards.-Visual standards for evaluation of metal defects will be established for each contract after the first months production. However, the establishment of visual standards will not waive the requirement for compliance with 3.4.1 and 3.4.2 (see 6.2).

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3.3.4 Base adapter.-The base adapter shall be free from cracks, splits, cold shuts, laminations, porosity or other metal defects.

3.4 Brazed joint (Body Assembly Alternative I)

3.4.1 Braze metal.-The braze metal shall be visible for the full 360 degrees where the base adapter joins the body wall on the inside diameter. The braze metal shall also be visible for the 360 degrees around the joint between the rear of the body and the base adapter flange. Voids will be allowed on the exterior surface only of the joint, providing, no single void is longer than 1/4 of an inch in length and the accumulated length of voids shall not exceed 1/2 inch of the outside periphery. Braze metal must be visible in all voids. Braze metal color variance shall not be cause for rejection.

3.4.2 Hydrostatic test.-The body assembly shall withstand the stresses developed by the application of 5000 pounds per square inch (psi) minimum, internal hydrostatic pressure for five seconds without leakage, rupture, or permanent deformation.

3.4.3 Air pressure test.-After hydrostatic testing, the body assembly shall withstand an internal air pressure of 150 psi minimum for 15 seconds minimum without evidence of leakage.

3.4.4 Joint quality.-The brazed joint shall be free of entrapped slag, voids, unwetted or unbonded areas and other brazing defects. The aggregate area of these defects shall not exceed 25 percent of the area of the faying surfaces; and the maximum extent of a defect including solder ring groove, shall not exceed 50 percent of the overlap distance (width) of the joint.

3.5 Protective coating.-Protective coating shall be in accordance with the applicable drawing and the specifications referenced thereon.

3.6 Proving Ground functional tests.-The projectile body assembly shall reveal no evidence of failure when tested at excess and service pressure. Body assembly failure shall be:

a. Projectile body and base adapter deformation attributable to noncompliance with drawing and specification requirements.

b. Loss or breakup of any component, except the obturator in the mortar tube or in flight.

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3.7 Workmanship.-The requirements for workmanship are as shown on the applicable drawings in the applicable specifications and the following:

3.7.1 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

3.7.2 Threads.-Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings and shall be in accordance with the provisions of MIL-A-2550.

3.7.3 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.7.4 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no grit, scale, accumulated braze alloy, residual flux or other foreign matter remains therein after final cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the Proving Ground functional tests (see 3.6).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground functional tests.

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4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 70 metal parts assemblies as follows:

55 projectile assemblies complete, painted
5 projectile assemblies complete, unpainted
10 sets of component parts, unassembled

Fifty of the above-painted projectile assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for the regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacture processes, material used, drawing, specifications or source of supply, such as to significally affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Heat treatment lots.-Heat treatment lots shall contain heat treated bodies from not more than one heat of steel that have been heat treated in the same heat treating equipment at the same temperature in one unchanged process. The heat treated lot thus constituted shall form the basis for selecting sample bodies for the tension test hereinafter prescribed.

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4.3.1.3 Stress relief lots.-Stress relief lots shall contain cold worked bodies from not more than one heat of steel that have been cold worked by the same process utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relief lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4 Miscellaneous.-Small lots, such as bodies from abnormally small heats of steel or bodies which have become separated from previously delivered lots, etc., may, at the discretion of the manufacturer, be grouped into miscellaneous heat treatment or stress relief lots as the case may be. Such lots shall be identified by a miscellaneous lot code number.

4.3.1.5 Grouping.-Heat treatment or stress relief lots shall remain grouped together for lot identity purposes, until such time as the results of the tension test have indicated that the lot has complied with the mechanical properties requirements of the applicable drawing.

4.3.1.6 Projectile lot.-All heat treatment or stress relief lots that have been determined to have complied with the mechanical properties requirements of the applicable component drawing, may be re-grouped for assembly into projectile lots. A projectile lot shall consist of projectile assemblies produced by one manufacturer, under one contract, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

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4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged at the beginning of each hours production from each machine in operation and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

4.3.2.2 Adapter, Base, Drawing D10535863 covering a detail of Drawing D10551951.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of brazing surface	Gage
102.	Diameter of shoulder	Gage
103.	Thickness through base	Gage
MINOR		
201.	Thickness of shoulder (min.)	Gage
202.	Length, total	Gage
203.	Radii or chamfers missing or incorrect	Visual (see 6.1)
204.	Surface finish improper	Visual (see 6.1)

4.3.2.3 Cover, Base, Drawing B7549017 covering a detail of Drawing D10551951.

CRITICAL		
	None defined	
MAJOR		
101.	Thickness	Gage
MINOR		
	None defined	

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4.3.2.4 Body Alternative I Drawing F10543032, covering a detail of Drawing D10551951.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Length, total	Gage
202. Diameter at rear of body	Gage

4.3.2.5 Body, Drawing F10543029 covering a detail of Drawing D10551951.

CRITICAL	
None defined	
MAJOR	
101. Thickness through base	Gage
MINOR	
None defined	

4.3.2.6 Projectile Assembly (prior to paint), Drawing D10551951.

CRITICAL	
1. Braze metal not visible for full 360 degrees at juncture of base adapter and body wall (Body Assembly Alternative I)	Visual
2. Hydrostatic test stamp missing (indicating failure to pass hydrostatic test) (Body Assembly Alternative I)	Visual
3. Air pressure stamp missing (indicating failure to pass air test) (Body Assembly Alternative I).	Visual
4. Metal defective	Visual (probe light)

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4.3.2.6 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Weight	Approved Scale
102.	Concentricity of pitch diameter of nose thread with bourrelet, including perpendicularity of front face	Gage
103.	Concentricity of band groove with bourrelet	Gage
104.	Concentricity at rear of body with bourrelet	Gage
105.	Concentricity of ogive with bourrelet	Gage
106.	Concentricity of pitch diameter of rear thread with bourrelet, including perpendicularity of rear shoulder	Gage
107.	Diameter of band groove	Gage
108.	Diameter of bourrelet, min.	Gage
109.	Thickness of wall at bourrelet	Gage
110.	Thickness of wall between bourrelet and taper, min.	Gage
111.	Thickness of wall at front of taper	Gage
112.	Thickness of wall at rear of taper	Gage
113.	Thickness through base at juncture of thread boss and rear face, min.	Gage
114.	Width of band groove	Gage
115.	Base cover weld or braze, inadequate	Manual (see 4.4.5)
116.	Surface finish of cavity, improper	Visual (see 6.1)
117.	Foregin matter in cavity	Visual
MINOR		
201.	Diameter of nose, outside	Gage
202.	Diameter at ogive basic length	Gage
203.	Diameter at rear bourrelet basic length	Gage
204.	Diameter at basic length on taper	Gage
205.	Diameter at rear basic length on taper (one piece design)	Gage
206.	Diameter of base thread undercut	Gage
207.	Diameter of base cover	Gage
208.	Diameter of base cover, min. (if applicable)	Gage
209.	Length of thread boss (including base cover)	Gage
210.	Length from front face to rear of shoulder (one piece design)	Gage

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
211. Length of nose thread cavity	Gage
212. Length, total, max.	Gage
213. Location of band groove	Gage
214. Radii or chamfers missing or incorrect	Visual (see 6.1)
215. Surface finish improper (except as otherwise specified)	Visual (see 6.1)
216. Overlap between base adapter and body (when applicable)	Visual
217. Burr	Visual
218. Foreign matter (except as otherwise specified)	Visual

4.3.2.7 Projectile Assembly (after paint) Drawing D10551951.

CRITICAL	
1. Metal defective	Visual (Probe light)
MAJOR	
101. Pitch diameter of nose thread, max.	Gage
102. Minor diameter of nose thread, max.	Gage
103. Pitch diameter of base thread, min.	Gage
104. Major diameter of base thread, min.	Gage
105. Pitch diameter of nose thread, min.	Gage
106. Minor diameter of nose thread, min.	Gage
107. Pitch diameter of base thread, max.	Gage
108. Major diameter of base thread, max.	Gage
109. Diameter of bourrelet, max.	Gage
110. Length of effective nose thread, min.	Gage
111. Length of effective rear thread, min.	Gage
112. Paint in cavity inadequate	Visual
113. Pool of paint in cavity	Visual
114. Thread(s) damaged or not full	Visual
115. Foreign matter	Visual
MINOR	
201. Corrosion preventive compound missing from threads and indicated surfaces	Visual
202. Paint on thread(s)	Visual
203. Paint not dry	Visual/Tactile
204. Paint inadequate or damaged	Visual
205. Marking missing, incorrect or illegible	Visual

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4.3.2.8 Packed Carton (prior to sealing).

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Support missing or damaged	Visual
202. Separator missing or damaged	Visual
203. Improper assembly	Visual
204. Incorrect number in carton	Visual

4.3.2.9 Packed Carton (after sealing).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton not completely sealed	Visual
203. Contents loose	Visual/Manual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness test.

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4.3.3.1.1.1 Heat treatment lots.-In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.3.1.2) shall be subjected to a hardness test with a Brinell Hardness Tester or by any other method of hardness testing approved by the procuring activity.

4.3.3.1.1.2 Stress relief lots.-Twenty-five stress relieved bodies from the first batch of each stress relief lot (as defined by 4.3.1.3) processed through the stress relief furnace and five bodies from each hour's production thereafter for the balance of the lot shall be subjected to a hardness test with a Rockwell Hardness Tester or by any other method approved by the procuring activity, to assess uniformity of the stress relief furnace. Hardness values shall be recorded.

4.3.3.1.1.3 Miscellaneous.-Each miscellaneous lot shall be subjected to the hardness tests listed in paragraphs 4.3.3.1.1.1 or 4.3.3.1.1.2, as applicable.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.-Two bodies, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier a production control sample may be utilized to establish the hardness range. When a production control sample is used the hardness range thus established shall be maintained throughout the lot. The size of the sample shall be determined by the supplier. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.3.3.1.2.2 Stress relieved lots.-Two bodies, one of the hardest and one of the softest as determined by the hardness test from the first twenty-five bodies selected in accordance with 4.3.3.1.1.2 shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirements of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout the stress relieved lot. The furnace temperature may vary within twenty-five degrees F. but not below the minimum temperature of 700 degrees F. and the soak time of 30 minutes minimum. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated every eight hours by an independent method.

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4.3.3.1.2.3 Miscellaneous lots.-Samples for the tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.3.3.1.2.1 or 4.3.3.1.2.2, as applicable. Failure of any tension test specimen to comply with the mechanical properties requirements of the applicable drawing, shall be cause for rejection of the lot. Double the number of samples shall be selected for miscellaneous lots.

4.3.3.1.3 Hardness pattern.

4.3.3.1.3.1 Heat treatment or stress relieved lots.-The samples supplied in accordance with 4.3.3.1.2.1 or 4.3.3.1.2.2, as applicable, shall also have a hardness pattern obtained to determine the hardest and softest section within the projectile and the basis for selecting the tension test specimens hereinafter prescribed. Failure to comply with the mechanical properties of the applicable drawing shall be cause for rejection of the lot.

4.3.3.2 Magnetic particle inspection.

4.3.3.2.1 After final heat treatment or stress relief and finish machining, bodies shall be inspected by magnetic particle inspection techniques approved by the responsible technical agency according to the following sequence.

(a) 100 percent magnetic particle inspection will be in affect until 2500 consecutive bodies have been inspected and found to be free of metal defects.

(b) Reduced inspection may then be instituted for magnetic particle inspection consisting of 30 bodies per hour for ten consecutive working days.

(c) If no defect is found during the ten days, the sample may be reduced to 15 bodies per hour for five consecutive working days.

(d) If no defect is encountered during the five days, the sample may be reduced to 5 bodies per hour.

(e) If a defect is encountered by magnetic particle inspection during reduced sampling (b through d) the magnetic particle inspection shall revert back to (a) above. In addition a 100 percent back screening to the last acceptable hours production will be performed.

(f) For requalification of reduced sampling, the manufacturer must follow the same procedures as outlined in paragraphs (a through d) above.

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4.3.3.2.1 (Cont'd)

(g) Should a controversy arise as to the validity of an indication, the manufacturer shall use metallographic test methods or an approved non-destructive method for proof of acceptability to the responsible technical agency.

(h) Continual application of a 100 percent magnetic particle inspection by paragraph (a) above, shall not be used as an alternative procedure in lieu of corrective manufacturing process action to eliminate a continuing process enduced defect. If corrective action is not instituted and the cause eliminated, the Government reserves the right to withdraw acceptance inspection until contractor can prove his process is capable of producing suspect free projectiles.

4.3.3.3 Brazed joint (Body Assembly Alternative I)

4.3.3.3.1 Braze metal.-Each body assembly shall be visually examined to determine compliance with 3.4.1. Failure to comply shall be cause for rejection of the body assembly.

4.3.3.3.2 Hydrostatic test.-Prior to gaging, each body assembly shall be subjected to an internal hydrostatic pressure to determine compliance with 3.4.2. Failure of any assembly to comply shall be cause for rejection of the assembly. Body assemblies which comply shall be stamped with the letter "H" as indicated on the assembly drawing.

4.3.3.3.3 Air pressure test.-Each body assembly that has passed the hydrostatic pressure test as indicated by the letter "H" stamped on the body, shall be subjected to a submerged air pressure test to determine compliance with 3.4.3. Failure of any assembly to comply as evidenced by the presence of air bubbles in the vicinity of the brazed joint or rising to the surface of the water shall be cause for rejection of the assembly. Body assemblies which comply shall immediately upon removal from the test fixture be stamped with the letter "A" as indicated on the assembly drawing.

4.3.3.3.4 Joint quality.-After air and hydrostatic testing but prior to the application of protective coating, 2 body assemblies from each lot shall be selected at random, sectioned and separated to determine compliance with paragraph 3.4.4. Failure of any specimen to comply shall be cause for rejection of the lot.

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4.3.3.4 Protective coating.-The sample size and acceptance criteria shall be in accordance with the assembly drawing and the requirements of the specifications listed thereon. Failure to comply with the requirements of the applicable specifications shall be cause for taking the corrective action recommended therein.

4.3.4 Proving Ground functional tests.-The following Proving Ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess pressure	25
Service pressure	25

4.3.4.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
Excess pressure	10 each lot
Service pressure	10 each lot

4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4 Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.5 Inspection equipment.-Index of Inspection Equipment List Number IEL-10551951 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation specified on the applicable IEL.

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4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness test.-Prior to finish machining, all bodies that will be subjected to the hardness test shall be prepared by grinding a flat on the cylindrical portion of the body where the obturator band groove will subsequently be machined. The test shall be performed using the equipment and procedures prescribed by Federal Test Method Standard No. 151, except that the minimum dwell time shall be 5 seconds when a Brinell Hardness Test is used.

4.4.1.2 Tension test.-From each sample selected in accordance with 4.3.3.1.2, four tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from opposite sides of the body, 180 degrees apart. The number of specimens and their location are for the convenience of the contractor or Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile meet the drawing requirements. Procedures and criteria for determining yield strength, elongation, reduction in area and tensile strength shall be as prescribed by Federal Test Method Standard No. 151, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0).

4.4.1.2.1 Hardness pattern.-From the bodies selected for the tension test, a hardness pattern shall be obtained by removing a longitudinal section, at least one-half inch thick extending the length of the projectile body. The section shall be removed from approximately the center of the projectile where the centerline of the specimen is with the walls of the specimen. The longitudinal walls shall be ground flat and parallel. The hardness test shall be performed on one side of the ground surface 1/4 of an inch apart along the longitudinal axis (see Figure I). Tension test specimens shall be taken from the hardest and softest sections of the body as indicated by the hardness pattern. In cases where tension test specimens cannot be removed from the hardest and softest areas because of the geometry of the projectile or shape of the hardness pattern, such as, transition zone, the matter shall be referred to the responsible technical agency for evaluation.

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4.4.2 Magnetic particle inspection.-The procedures for performing this test utilizing the wet fluorescent method shall be as specified in MIL-I-6868, for indirect magnetization by either the continuous method or the residual method, utilizing a circular magnetic field. The suspension shall be applied to the entire body, inside and outside. After draining, the body shall be thoroughly inspected in a darkened area under an ultra-violet light which will produce adequate fluorescence of the magnetic particles. The solid content of the suspension shall be maintained at 0.15 to 0.4 milliliter (ml) of solids per 100 ml of suspension. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by specification MIL-I-6868. When performing the test for solid content of the suspension, if it is determined that it does not comply, all projectile bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. After acceptance the body shall be demagnetized and cleaned. The magnetic particle inspection shall be performed on equipment that complies with the requirements of PA-PD-792. The operator(s) of the equipment and inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

4.4.3 Brazed joint (Body Assembly Alternative I)

4.4.3.1 Braze metal.-Visual examination shall be performed by using a four power (4X) magnification and a light source.

4.4.3.2 Hydrostatic test.-The body assembly shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached, after which the pressure will be held for a minimum of 5 seconds. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 2 hours of continuous operation. It shall also be calibrated prior to re-starting the operation after a layoff of one hour or more.

4.4.3.3 Air pressure test.-This test shall be made using equipment capable of performing the test accurately and consistently. Body assemblies shall be mounted in an approved fixture in such a manner that the body assembly shall be completely submerged in water; the required air pressure shall be applied and held for a minimum of 15 seconds. The water in the tank shall be permitted to settle after immersion of the assemblies prior to the start of timing. Observation for evidence of leakage shall be made for the entire time the assembly is required to be pressurized. The equipment shall be calibrated at the start of each day's operation and at intervals of 4 hours and after a stoppage of 3 hours or more.

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4.4.3.4 Joint quality.-The samples selected in accordance with paragraph 4.3.3.3.4 shall be sectioned and separated as follows: (a) the rear portion of the projectile assembly, 1-inch minimum forward of the base adapter, shall be removed in such a manner that no portion of the brazed joint will be damaged, (b) the rear portion of the body containing the base adapter shall then be cut vertically through the center of the base adapter, (c) each portion of the body shall be peeled from the base adapter section. After peeling, the brazed area of the two halves of the base adapter and the mating pieces of the body shall be visually examined to determine compliance with 3.4.4.

4.4.4 Protective coating.-The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specifications listed thereon. The salt spray test shall be performed utilizing equipment prescribed by Federal Test Method Standard No. 141.

4.4.5 Base cover weld or braze.-The test of the base cover weld or braze shall be performed as follows: Visually inspect the base shield and select the area that appears to have the weakest weld; insert a 5/8" chisel in this area and apply force by use of a 1-pound steel hammer at an approximate 1-foot stroke. If the weld breaks loose after 2 strokes, the body assembly will be classed defective. If the weld is found to be secure, the burr made by the chisel may be knocked back into position and accepted.

4.4.6 Proving Ground functional tests.

4.4.6.1 Excess pressure.-The samples supplied in accordance with Tables I and II inert loaded to the service weight and dummy fuzed will be assembled into complete rounds and fired for recovery in a mortar for which the round is standard. The propellant charge will be adjusted to produce 112 percent of the maximum rated pressure of the round in a new mortar.

4.4.6.2 Service pressure.-The samples supplied in accordance with Tables I and II, inert loaded and fuzed will be assembled into complete rounds utilizing the service charge and fired in a mortar for which the round is standard for comparison with an equal number of rounds from a reference lot. The sample round and reference rounds will be fired alternately. Horizontal and vertical distances will be recorded for each round fired for determination of horizontal and vertical probable error (PE) in mils to assess accuracy. Data obtained will be recorded for information only and forwarded to the cognizant technical agency.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Packaging, Packing and Marking shall be in accordance with the requirements of level C in MIL-STD-1169.

5.2 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards to be established at contractors plant is accordance with 3.3.3.

6.3 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing activity

Army - MU(FA)

Project No. 1315-A726

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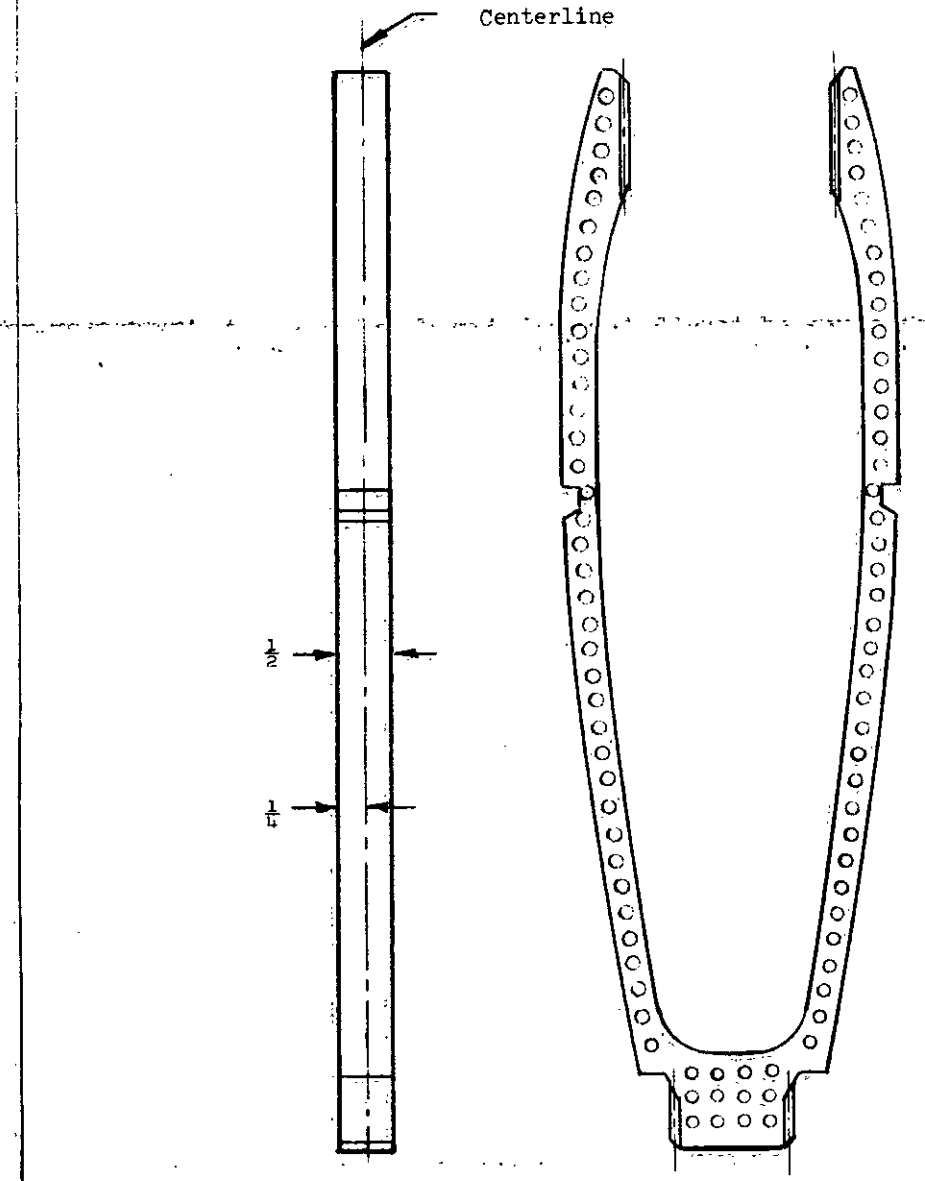


FIGURE I

Hardness Pattern

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POSTAGE AND FEES PAID
Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-T1000
Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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